



GCOM-C Guidelines

JAXA/EORC
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v2026.5.8



GCOM-C Guidelines

GCOM-C Guidelines List

GCOM-C /Hardware

Orbit

Contents	URL
GCOM-C orbit map (images, ftp)	https://suzaku.eorc.jaxa.jp/GCOM_C/orbit/orbit_map
ECGPM,GCOM-C predicted orbit map (Google map)	https://www.eorc.jaxa.jp/EARTH/CARE/auth/orbit/
ECGPM,GCOM-C orbit map search (Google map)	https://www.eorc.jaxa.jp/EARTH/CARE/auth/orbit/

Sensor

Sensor specification (eo-portal)	https://www.eoportal.org/satellite-missions/gcom-c
SGLI Sensor Characterization (Spectral Response Function, etc.)	https://suzaku.eorc.jaxa.jp/GCOM_C/data/prelaunch
GCOM-C post launch calibration (vicarious calibration coefficient, etc.)	https://suzaku.eorc.jaxa.jp/GCOM_C/data/prelaunch
SGLI Calibration Event List	https://shikisai.jaxa.jp/docs/event_information.pdf
SGLI Missing Data List	https://shikisai.jaxa.jp/docs/SGLI_data_missing.pdf

Instruments

JAXA's In-situ instruments availability	https://docs.google.com/spreadsheets/d/1BChSu6F
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GCOM-C /Software

Product

Products information: stored physical quantity, ATBD	https://suzaku.eorc.jaxa.jp/GCOM_C/data/product
How to use QA_flag	https://shikisai.jaxa.jp/faq/faq0040_j.html
SGLI SGLI Level-1 Product Format Manual (レベル1 プロダクトフォーマット説明書)	https://eoportal.jaxa.jp/gpr/assets/mng_upload/GCOM-C
SGLI Level-2 & 3 Product Format Manual (高次プロダクトフォーマット説明書)	https://eoportal.jaxa.jp/gpr/assets/mng_upload/GCOM-C

Data DL

Data File Naming rules, Products list, TILE definition, Data structure, HDF5 data loading and processing method: HDF5データ読み込み・処理方法, 命名規則, プロダクト一覧, タイル定義, データ構造, etc.)	https://shikisai.jaxa.jp/faq/docs/GCOM-C_Product
Data File Naming rules	https://shikisai.jaxa.jp/faq/docs/GCOM-C_FAQ_GI
[h5/geotiff] G-Portal (can DL the JAXA satellites standard products)	https://eoportal.jaxa.jp/gpr/?lang=ja
[h5/geotiff] How to DL from G-Portal (Youtube)	https://www.youtube.com/watch?v=Q8dvRlet2Y
[netCDF] JASMES dataset (can DL the netCDF shown in JASMES, including the GCOM-C research products)	https://www.eorc.jaxa.jp/JASMES/registration.html
G-portal User Manual	https://earth.jaxa.jp/gpr/ToolsDocs/docs/GPortalU

[python] How to get File name that corresponds to the area you are interested in	
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How to analyze

[QGIS] G-Portal→L2-LST DL(h5→geoTiff)→show in QGIS	https://eoportal.jaxa.jp/gpr/assets/mng_upload/CO
[QGIS] G-Portal→L2-SST DL(h5→geoTiff)→show in QGIS (Youtube)	https://www.youtube.com/watch?v=9hixQL72K_U
[SeaDAS] JASMES→netCDF DL→show in SeaDAS	https://www.eorc.jaxa.jp/JASMES/SGLI_NRT/JASM
[convert/QGIS] JASMES SGLI, VIIRS conversion from netCDF to GeoTIFF→show in QGIS	https://www.eorc.jaxa.jp/JASMES/tutorial/tutorial_
[convert/QGIS] JASMES MODIS conversion from binary to GeoTIFF→show in QGIS	https://www.eorc.jaxa.jp/JASMES/tutorial/tutorial_
[convert/QGIS] JASMES MODIS conversion from binary to netCDF→show in QGIS	https://www.eorc.jaxa.jp/JASMES/tutorial/tutorial_
[python/c++]-sample code: G-Portal→h5 DL→L2 Scene/Tile map projection→show in python/c++	https://shikisai.jaxa.jp/faq/faq0069_j.html
[python] SGLI Python Open Tool (SPOT): G-Portal→h5 DL→L1/L2 Scene/Tile map projection→show in p	https://github.com/K0gata/SGLI_Python_Open_To

JASMES

JASMES Top	https://www.eorc.jaxa.jp/JASMES/index.html
Image Archive (Global, Japan)	https://www.eorc.jaxa.jp/JASMES/Image/archive_j
Image Analyzer (can scroll, analyze temporal variation, etc.)	https://www.eorc.jaxa.jp/JASMES/Image/analyzer_j
SGLI Near-Realtime Monitor	https://www.eorc.jaxa.jp/JASMES/SGLI_NRT/index
Greenland monitor	https://www.eorc.jaxa.jp/JASMES/daily/GLmonitor
How to DL from JASMES (FTP)	https://www.eorc.jaxa.jp/JASMES/registration_j.htm
	https://www.eorc.jaxa.jp/JASMES/docs/JASMES_D

JAXA API

Google Earth Engine (Around Japan: RGB, LST, CHLA, TSM, SST, FAI, RSRF)	https://shikisai.jaxa.jp/GEE/DataList/index_j.html
Google Earth Engine Instruction Top	https://shikisai.jaxa.jp/GEE/index_j.html

JAXA Dashboard

JAXA API	https://data.earth.jaxa.jp/ja/
QGIS API Plug-in	https://data.earth.jaxa.jp/eop/ggis/ja/
Python API	https://data.earth.jaxa.jp/eop/python/v0.1.4/en/goo

You can access the left excel



Visualization

Python Tool to get SGLI file via sftp

- ✓ **A Python tool to get the SGLI file from G-Portal for any L1/L2/L3-product_ID and any latitude/longitude range and any date.**

Usage:

```
python get_sgli_granuleID_and_file.py <quantity> <orbit> <date_yyyymmdd>  
<lat_ll> <lon_ll> <lat_ur> <lon_ur> [output_dir] [--auto]
```

Examples:

```
python get_sgli_granuleID_and_file.py SST D 20250702 35 135 45 150  
python get_sgli_granuleID_and_file.py NDVIF D 20250701 0 0 0 0 ./download  
python get_sgli_granuleID_and_file.py IWPRK D 20210710 35.0 139.5 35.8  
140.2 ./download  
python get_sgli_granuleID_and_file.py SST_Q D 20250702 35 135 45  
150 ./download --auto
```

Arguments:

<quantity>	Quantity code (e.g. SST, SST_Q, NDVIF, CHL)
<orbit>	Orbit direction: A=Ascending / D=Descending
<date_yyyymmdd>	Target date (e.g. 20250702)
<lat_ll>	Bounding box lower latitude (degrees)
<lon_ll>	Bounding box lower longitude (degrees)
<lat_ur>	Bounding box upper latitude (degrees)
<lon_ur>	Bounding box upper longitude (degrees)
[output_dir]	Output directory, defaults to the current directory
[--auto]	Skip the confirmation prompt and start downloading automatically

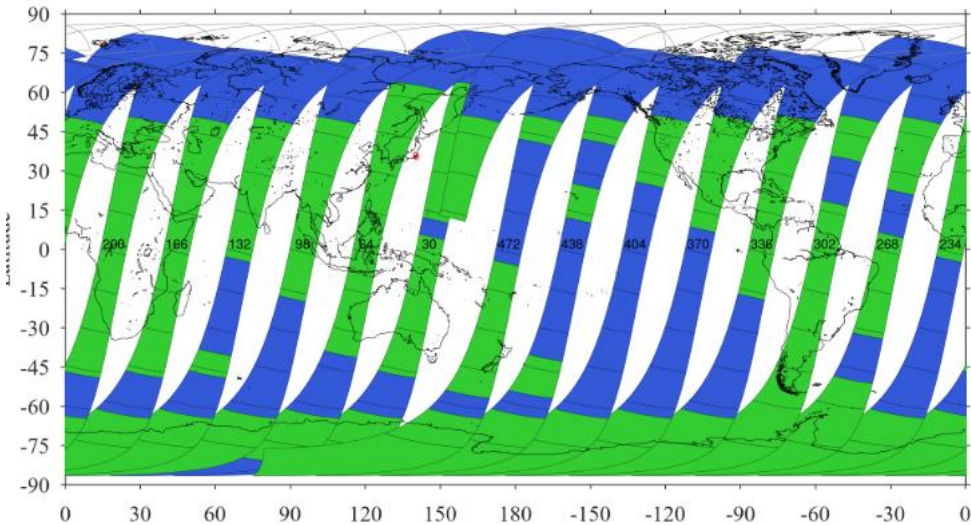
Github



GCOM-C orbit map

GCOM-C orbit map

Date: 20250101 (REQ: REQEORC5A20250101_p0)



Descending North pole



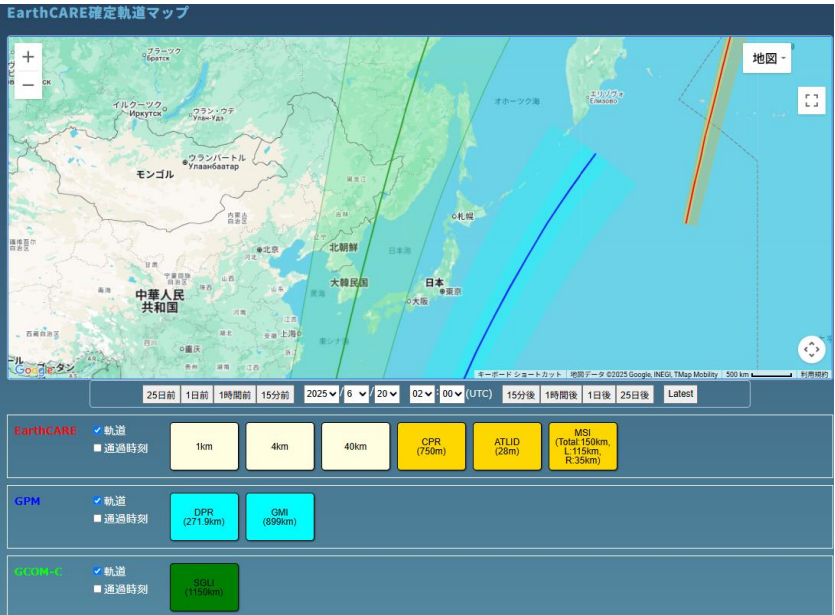
Descending South pole



- Past and future observation plans of GCOM-C.

https://suzaku.eorc.jaxa.jp/GCOM_C/orbit/orbit_map_j.html

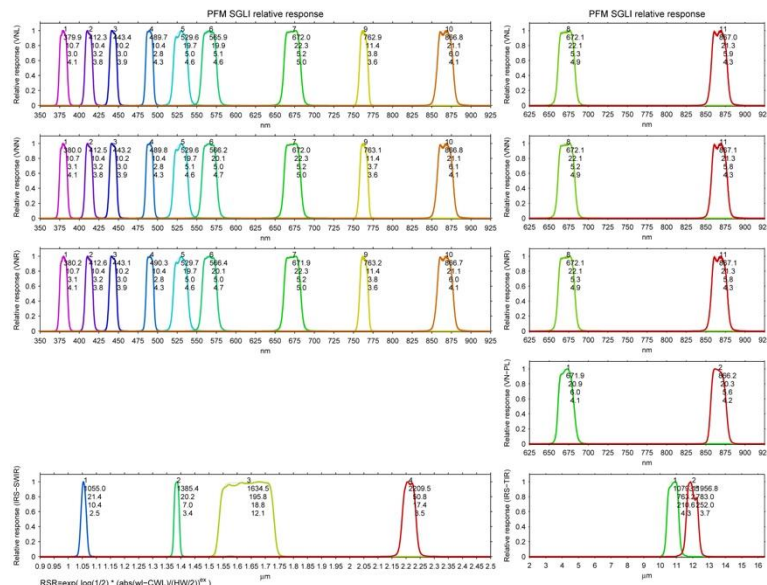
EC,GPM,GCOM-C orbit map search (Google map)



- The orbits of EC, GPM, and GCOM-C on Google Map

https://www.eorc.jaxa.jp/EARTH/CARE/auth/orbit/rs/EC_orbit_search_j.html

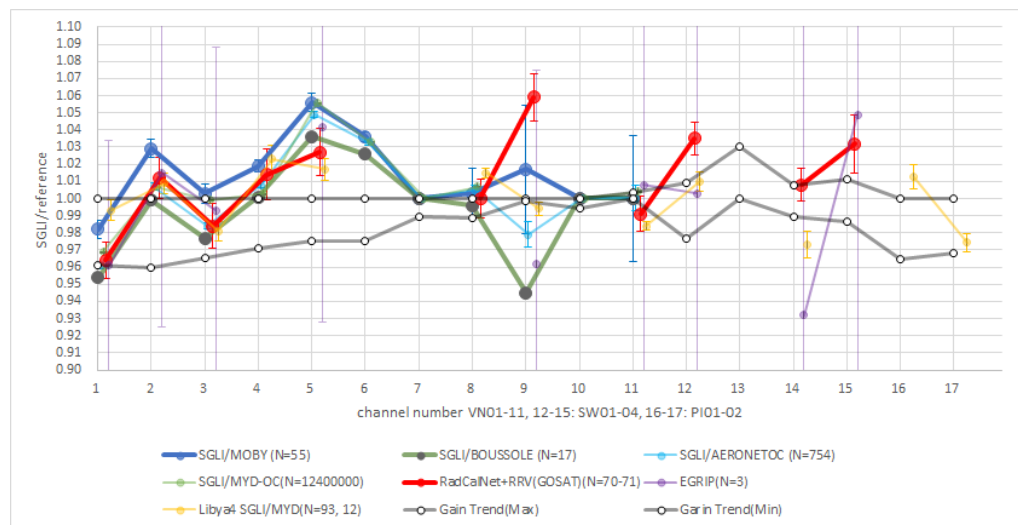
SGLI Sensor Characterization



- Spectral Response Function, etc.

https://suzaku.eorc.jaxa.jp/GCOM_C/data/prelaunch/index.html

GCOM-C post launch calibration



- Vicarious calibration coefficient, etc.

https://suzaku.eorc.jaxa.jp/GCOM_C/data/prelaunch/index_cal.html

Products information: stored physical quantity, ATBD

HOME	About GCOM-C	Instruments	Data Products	Resources	Earth Monitoring
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"Product ID" (You can download Algorithm Theoretical Basis Documents (ATBDs) and sample data). In addition, the definition of the data provided on G-Portal can be checked at the "Detail" button in "Category" column.

The map projected images (Japan area and global) of the standard products are provided at JASMES/SGLI Standard Monitor.

The map projected images (Japan area) of the near realtime products are provided at JASMES/SGLI Near-Realtime Monitor.

Ver.1	Ver.2	Ver.3				
Category	Product [ID*1]	Definition & Unit*2	Algorithm ID	Responsible Person	ATBD (update date)	Ver.3 update contents
Radiance	Level-1A [VNR/IRS/POL]	Def.: Sensor output (digital counts) Unit: none	G1	JAXA	-	• Not changed (Major version is continued as Ver.2. Minor version is updated.)
	Level-1B [VNR/IRS/POL]	Def.: Calibrated radiance (granule scene or path data projected on reference coordinate) Unit: W/m2/um/sr	G1	JAXA	-	• Not changed (Major version is continued as Ver.2. Minor version is updated.)
	LTOA (Tile)	Def.: Top of atmosphere (TOA) radiance Unit: W/m2/um/sr	G2A G2B G2C	JAXA Hashimoto Takaku	Ver.1 (Sep. 2020)	• Not changed (Major version is continued as Ver.2. Minor version is updated.)
	LTOA (Global)		G3A	H. Murakami	Ver.1 (Oct. 2018) Ver.3 (Oct. 2021)	• Modified pixel selection method
	LCLR (Global)	Def.: Top of atmosphere (TOA) radiance only for clear Unit: W/m2/um/sr				• Modified pixel selection method (Heavy aerosol flag of CLFG is used as clear sky)
Land	Atmospheric corrected reflectance [RSRF]	Def.: Land surface reflectance corrected for the effects of atmospheric scattering and absorption. Correction of directional anisotropic effects are also made for 8-day and monthly composite products. Unit: none	T1A	H. Murakami	Ver.1 (Oct. 2018) Ver.2 (Mar. 2020) Ver.3 (Oct. 2021)	• Modified coefficients of cloud, discrimination of shadow, and surface reflectance. • Added polarized reflectances (Rp_PL01, Rp_PL02) to Image_data. • Added limit of target procession days • Changed Error_DN of QA_flag to "1" in order to match with bit0=1(no data)
	Vegetation index [VGI_(NDVI/EVI_)]	Def.: Indices indicating vegetation cover and activity such as NDVI and EVI Unit: none	T2A	JAXA	Ver.1 (Nov. 2018)	• Added Cosine_of_solar_incidence from slope normal to Geometry_data • Modified QA_flag • Modified SDI calculation parameters
	Shadow index	Def.: An index indicating		M. Morioka	Ver.1 (Sep. 2014)	

https://suzaku.eorc.jaxa.jp/GCOM_C/data/product_std_j.html

GCOM-C
Global Change Observation Mission-Climate

PI's DOOR 日本語

HOMEAbout GCOM-CInstrumentsData ProductsResourcesEarth Monitoring

Top

Vegetation Index (VGI)

[DOI: 10.57746/E0.01gs73b3y17efdvjm1ajftbfr2]

- Introduction
- Update contents of Ver.2
- Update contents of Ver.3
- Sample image
- Dataset name
- QA_flag information
- Attribute information
- How to Use

Introduction		
NDVI, EVI	Define	Indices indicating vegetation cover and activity such as NDVI and EVI
	Unit	none
	ATBD	Ver.1 (Nov. 2018)
SDI	Define	An index indicating shadow fraction of vegetation area inferred from spectral reflectance.
	Unit	none
	ATBD	Ver.1 (Sep. 2014) Ver.2 (May 2020) Ver.3 (Oct. 2021)

QA_flag information					
Ver.1	Ver.2	Ver.3	Ver.1(Sample Image)	Ver.2(Sample Image)	Ver.3(Sample Image)
Bit	Data name	Description	EV1_L3_mask k*	NDVI_L3_mask k*	SDI_L3_mask k*
0	no data	1: VN08 and/or VN11 are error	1	1	1
1	land/water	0: water >50%, 1: land > 50%	0	0	0
2	mixed with land/water	1: land and water > 1%	0	0	0
3	cloud	1: cloud	1	1	1
4	probably cloud	1: probably cloud	0	0	0
5	snow or ice	1: snow or ice	0	0	0

Attribute information		
Ver.1	Ver.2,3	
Attribute		Remarks
Attribute of Image_data		
EVI		
NDVI	Error_DN	65535
16-bit unsigned integer	Minimum_valid_DN	0
	Maximum_valid_DN	20000
	Slope	0.0001
	Offset	-1
	Spatial_resolution	0.002083333250m: 0.002083333
	Data_description	Normalized Difference Vegetation Index (NDVI)
	Unit	NA
	Spatial_resolution_unit	degree
	Mask_for_statistics	1225
		Detail: QA_flag_information(~L3_mask)
SDI		

- ATBD
- QA_flag
- Sample image
- Attributes
- Updates info
- Etc.

GCOM-C/SGLI File name rule (L1/L2 Scene)

e.g.) GC1SG1_202105100122H05010_L2SG_IWPRQ_3000.h5

L2 scene

Byte	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	
GID	G	C	1	S	G	1	-	Y	Y	Y	Y	M	M	D	D	H	H	m	m	s	P	P	P	S	S	-	L	L	x1	x2	-	K	K	K	K	r	-	a	p	p	p	
Example	G	C	1	S	G	1	-	2	0	2	4	0	2	1	3	0	1	1	5	R	0	4	7	0	9	-	L	2	S	G	-	S	S	T	-	K	-	3	0	0	1	
Contents	Satellite (fix)		Sensor (fix)		-		Year				Month		Day		Hour		min		sec	Path※2		Scene ※3		-		Level ※8		Type ※9		-		Subsystem ※12		resolution ※13		-		algorithm ver.※14		parameter ver. ※15		
	Observation start UT※1																																									

※1
•Set the ideal start time of product. (UTC time)
•As there is some overlap before and after product, it does not match with the time of product start line.
The number of seconds is expressed by single alphabet.

※2 Path number : 1~485
※3 Scene number : 1~24

※8 Level : L2

※9 Type
SG : Standard processing (global)
SL : Near-real time processing (around Japan)
SN : Near-real time processing (global)

※12 物理量
OKID
SST_
IWPR
NWLR

※13 Resolution
Q : 250m
K : 1000m
L : 1000m(low resolution resampling)
IRS has other pattern. (H, Y, X, M)

※14 algorithm ver. : 0~9、A~Z
※15 parameter ver. : 000~999

L1 scene

32	33	34	35	36
K	K	K	m	r
V	N	R	D	Q
Subsystem ※10		Mode※11		resolution ※13

※10 Subsystem
VNR : VNR-NP
POL : VNR-PL
IRS : IRS (SWIR+TIR)

※11 Mode
D : Day
N : Night

GCOM-C/SGLI File name rule (L2/L3 Tile, Global, Statistics)

e.g.) GC1SG1_20210601D01M_T0620_L2SG_SWR_Q_3000.h5

L2/L3 Tile, Global, Statistics

Byte	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
GID	G	C	1	S	G	1	-	Y	Y	Y	Y	M	M	D	D	H	H	m	m	s	P	P	P	S	S	-	L	L	x1	x2	-	K	K	K	K	r	-	a	p	p	p
Example	G	C	1	S	G	1	-	2	0	2	4	0	2	1	3	0	1	1	5	R	0	4	7	0	9	-	L	2	S	G	-	S	S	T	-	K	-	3	0	0	1
Contents	Satellite		Sensor					Year			Month	Day	Hour	min	sec	Path※2		Scene	Level		Type	Subsystem			resolution		algorithm		parameter												
	(fix)		(fix)					Observation start UT※1								※3		※8	※9		※12			※13			ver.※14	ver.													
																																									※15

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SG : Standard processing (global)
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※14 algorithm ver. : 0~9, A~Z
※15 parameter ver. : 000~999

G-Portal

様々な分野で利用いただくことを目的とした地球観測データを無償で提供

G-Portal

[トップへ](#) | [初めての方へ](#) | [サポート](#) | [ログイン](#)

[保存した検索条件の呼び出し](#) [検索条件の保存](#)

[ガイダンスを表示](#)

省景地図の変更 [Google ストリート](#)

1.絞り込み 2.期間指定 3.範囲指定

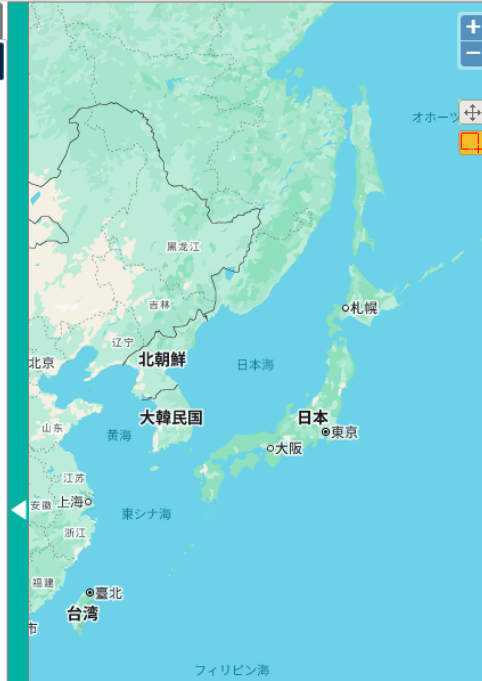
物理量から選ぶ 衛星、センサから選ぶ

1.絞り込み条件を設定します。

単語での絞り込み 絞り込み

処理レベル 機能 すべて

衛星、センサ、物理量	情報	設定
☒ GCOM-C/SGLI	i	
☐ LEVEL1		
☒ LEVEL2		
☐ 海洋圏		
☒ 雪氷圏		
☒ L2-オホーツク海氷分布	i	⚙
☐ L2-積雪・海氷分布	i	⚙
☐ L2-雪氷面温度&浅層積雪粒径&雪氷面アルベド	i	⚙
☐ L2-雪氷面温度&浅層積雪粒径 ver.2	i	⚙
☐ 雪氷圏統計		
☐ 陸圏		
☐ 陸圏統計		
☐ 大気圏		
☐ 大気圏全球		
☐ LEVEL3		
☐ GCOM-W/AMSR2	i	
☐ GPM	i	
☐ GPM Constellation satellites	i	
☐ GSMP	i	



<https://gportal.jaxa.jp/gpr/?lang=ja>

- SGLI Standard products
- Hdf5 format (can be converted to GeoTiff)
- 1/24 deg(or 1/12deg) and 250m
- Only JAXA's satellites data are available

JASMES

JASMES Data Access

Portal
TOP

EORC
GCOM-C

At
D

JASMES Data Access

1. User Registration

JASMES site provides various products in binary and netCDF formats.

Please read the [terms and conditions](#) carefully and agree on them before the use of the Service, and

If you want to change or delete the user registration information, please click [here](#).

Please check [here](#) for how to get JASMES data.

User Registration

2. About Data

The following JASMES site provides various products in binary and netCDF formats.

- 1) [JASMES Image Archive](#)
- 2) [SGLI Near-realtime Monitor Service](#)
- 3) [MODIS Near-realtime Monitor Service](#)
- 4) [JASMES Image Analyzer](#)
- 5) [FAI Monitor](#)

Please refer to [here](#) to see Data list.

To access GCOM-W1/AMSR2 data, please use [the Global Portal Service \(G-Portal\)](#).

<https://www.eorc.jaxa.jp/JASMES/registration.html>

- SGLI Standard and Research products
- Basically, netCDF format
- 5km for global, 250m for Japan
- VIIRS, MODIS are also available

How to analyze in a local environment

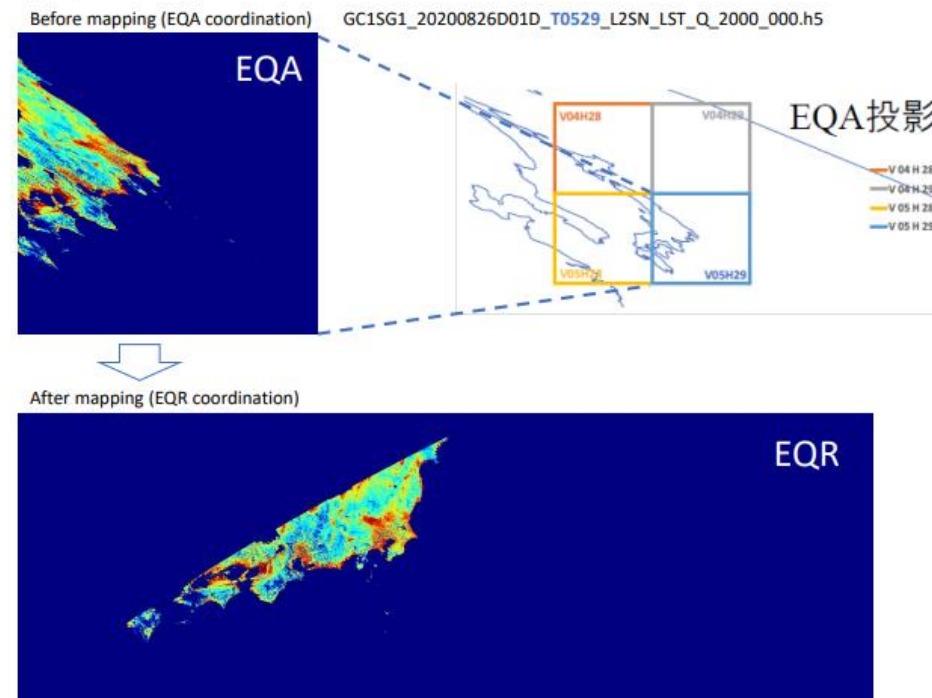
[QGIS] G-Portal→L2-LST
DL(h5→geoTiff)→show in QGIS



[https://gportal.jaxa.jp/gpr/assets/mng_upload/CO2MMON/upload/Imaging_procedure_for_Level-2_LST_product_of_SHIKISAI\(GCOM-C\)_using_QGIS.jp.pdf](https://gportal.jaxa.jp/gpr/assets/mng_upload/CO2MMON/upload/Imaging_procedure_for_Level-2_LST_product_of_SHIKISAI(GCOM-C)_using_QGIS.jp.pdf)

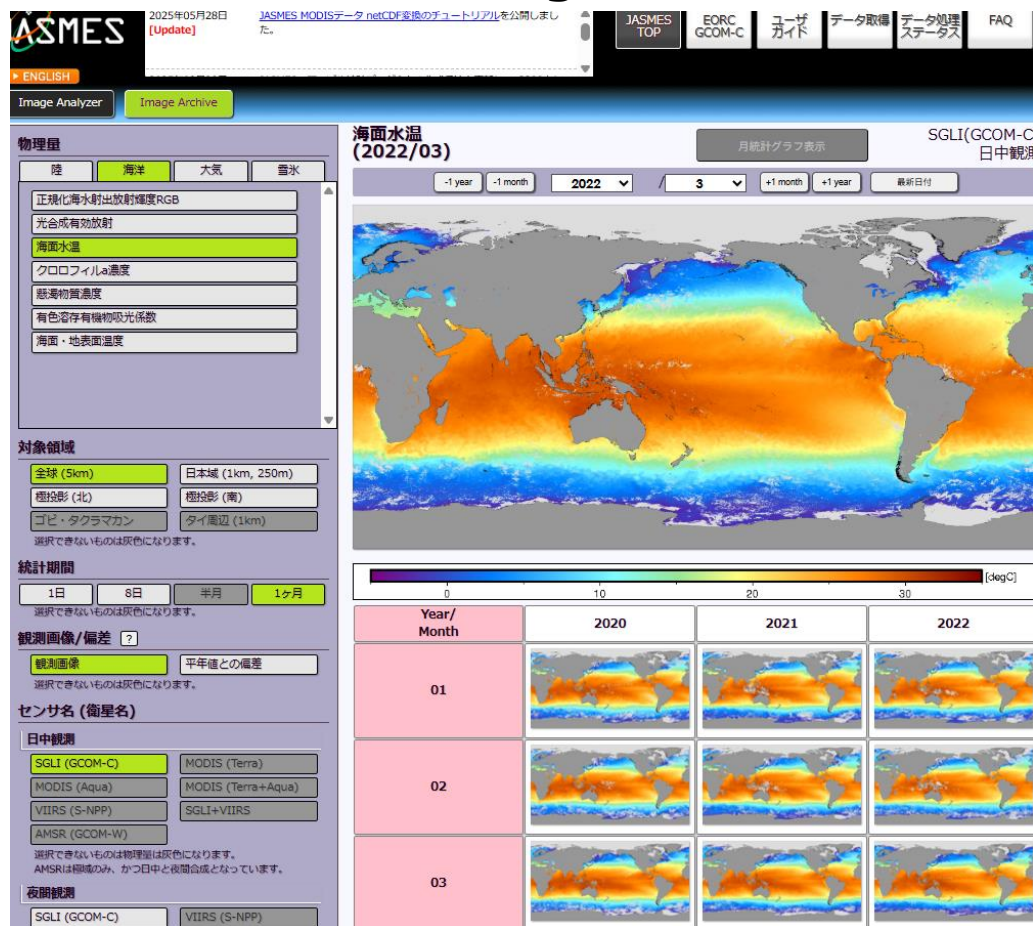
[python/c++] sample code: G-Portal→h5 DL→L2
Scene/Tile map projection→show in python/c++

Map projection Example



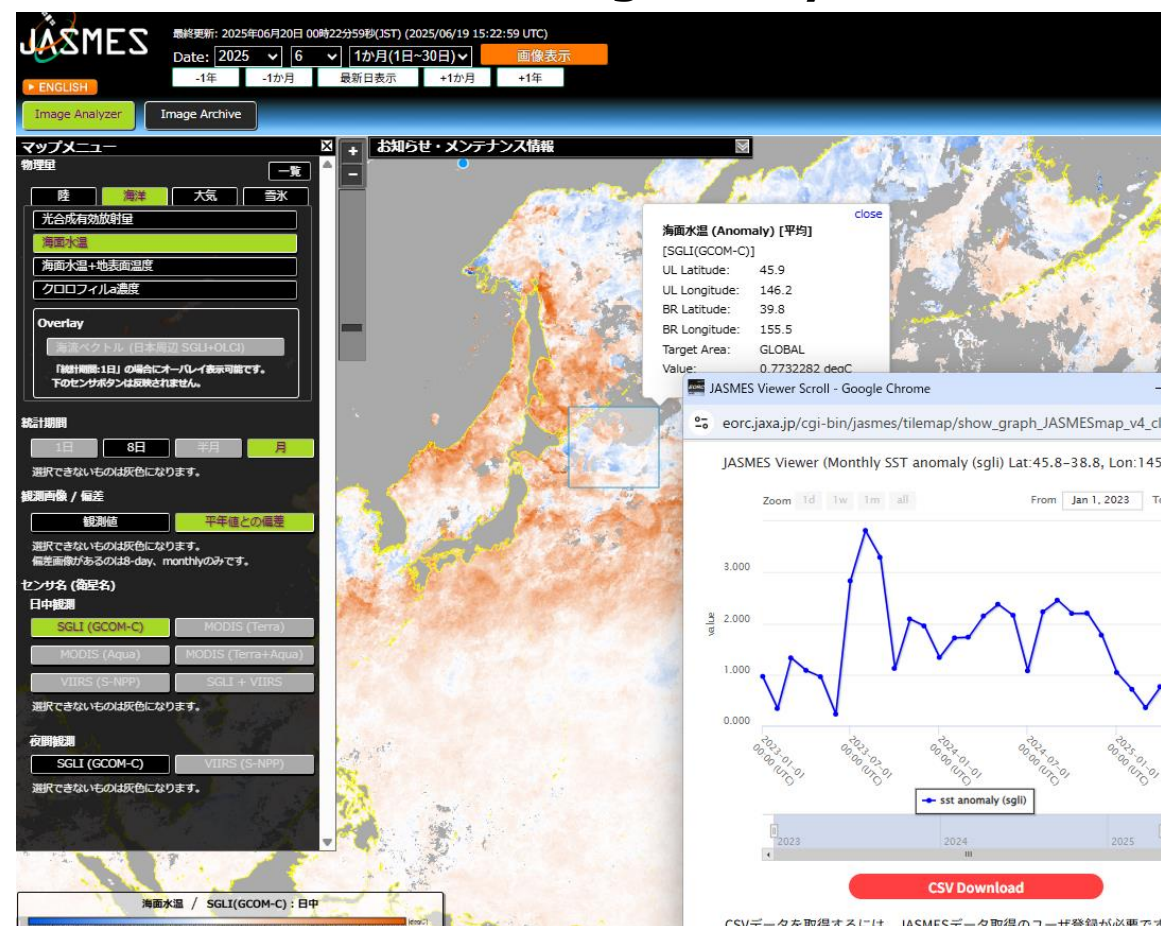
https://shikisai.jaxa.jp/faq/faq0069_j.html

JASMES Image Archive



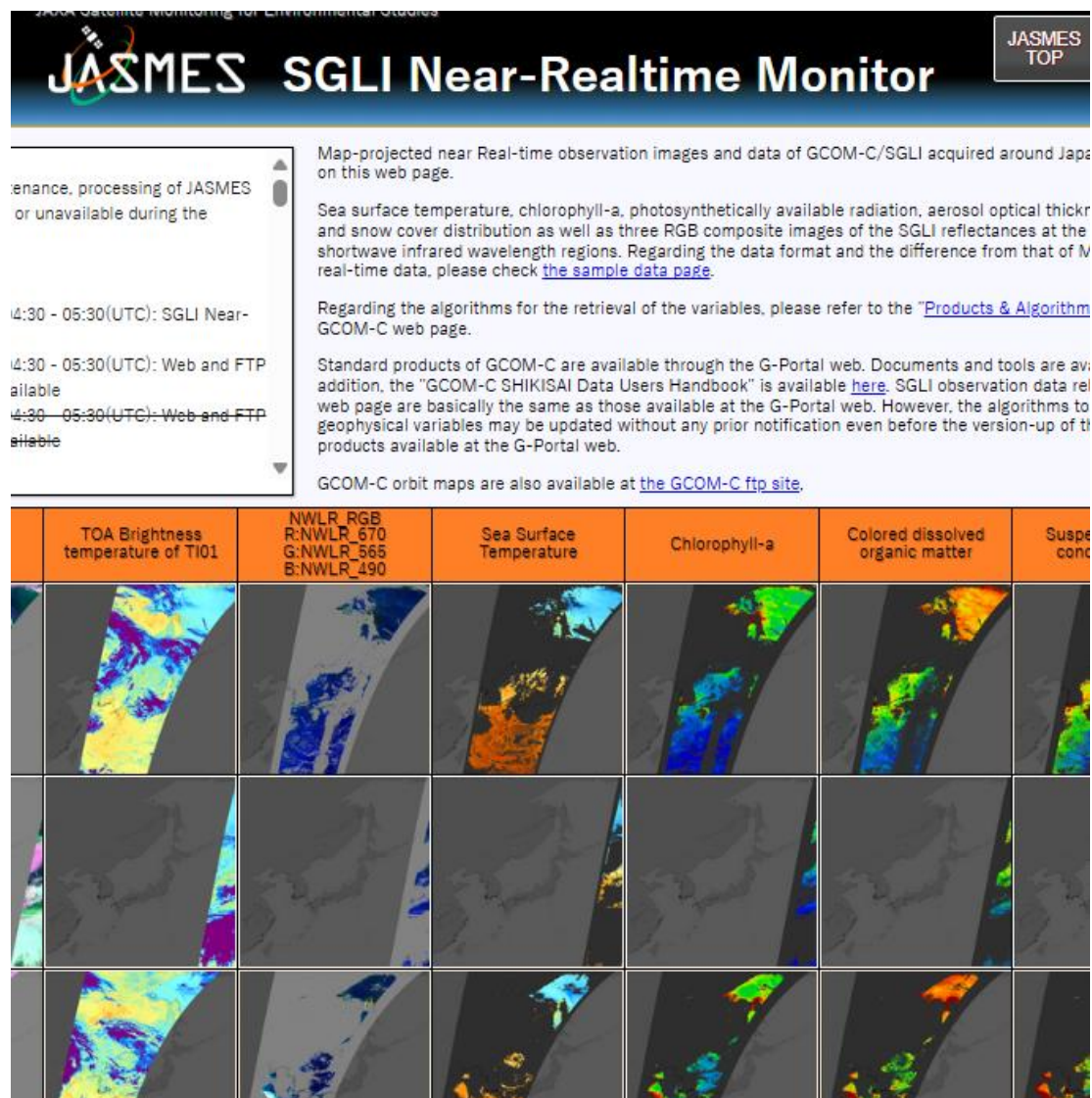
- Visualize the images like a calendar
- Global: 5km, Japan: 250m
- Absolute value and anomaly are available

JASMES Image Analyzer



- Visualize the images like a google map
- Global: 5km, Japan: 250m
- Absolute value and anomaly are available
- Time series analysis is available

JASMES Near-Realtime Monitor



- Visualize the Near-Realtime SGLI data
- Japan area is available
- If you would like to analyze long-term variation, you should use standard product (the accuracy of NRT data is worse than standard data)

JAXA Earth API (QGIS, Python, Java)



Usage examples for Python

- Get and show an image
- Search collection's ID and Band
- Get and show an image by region of interest
- Get and show a masked image
- Get and show a differential image
- Get and show a composite image
- Calculate and show timeseries data

Examples of using the API for Python in QGIS

- Getting data with Google Colab
- API Reference
- Database for API
- Release notes
- Documents
- COG/STAC Generator with Docker
- Access Analysis Tool with Metabase

Usage examples for Python

Get and show an image

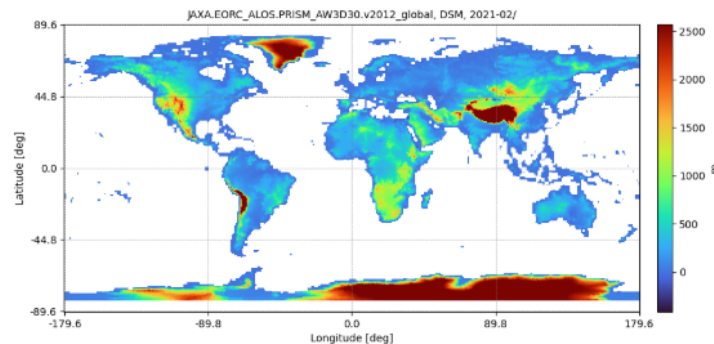
Following Python script is the simplest usage example of the API to get and show an image. Each method's default value is used if the user doesn't input as argument. If you get ssl error, please set `ssl_verify` to `False`.

```
# Load module
from jaxa.earth import je

# Get an image
data = je.ImageCollection(ssl_verify=True)\
    .filter_date()\
    .filter_resolution()\
    .filter_bounds()\
    .select()\
    .get_images()

# Process and show an image
img = je.ImageProcess(data)\
    .show_images()
```

A following image will be shown.



If you would like to get raw numpy array and show it, please execute following script.

- Get the JAXA's dataset (including JASMES data?) and analyze on QGIS, python, java

Search:

Aerosol Optical Depth at 500 nm (Daytime/Monthly)

Aerosol optical thickness product in GCOM-C SGLI Level-3 Standard product Version 3.

ID:

BAND:

Chlorophyll-a concentration (Daytime/Monthly)

Chlorophyll a concentration product in GCOM-C SGLI Level-3 Standard product Version 3.

ID:

BAND:

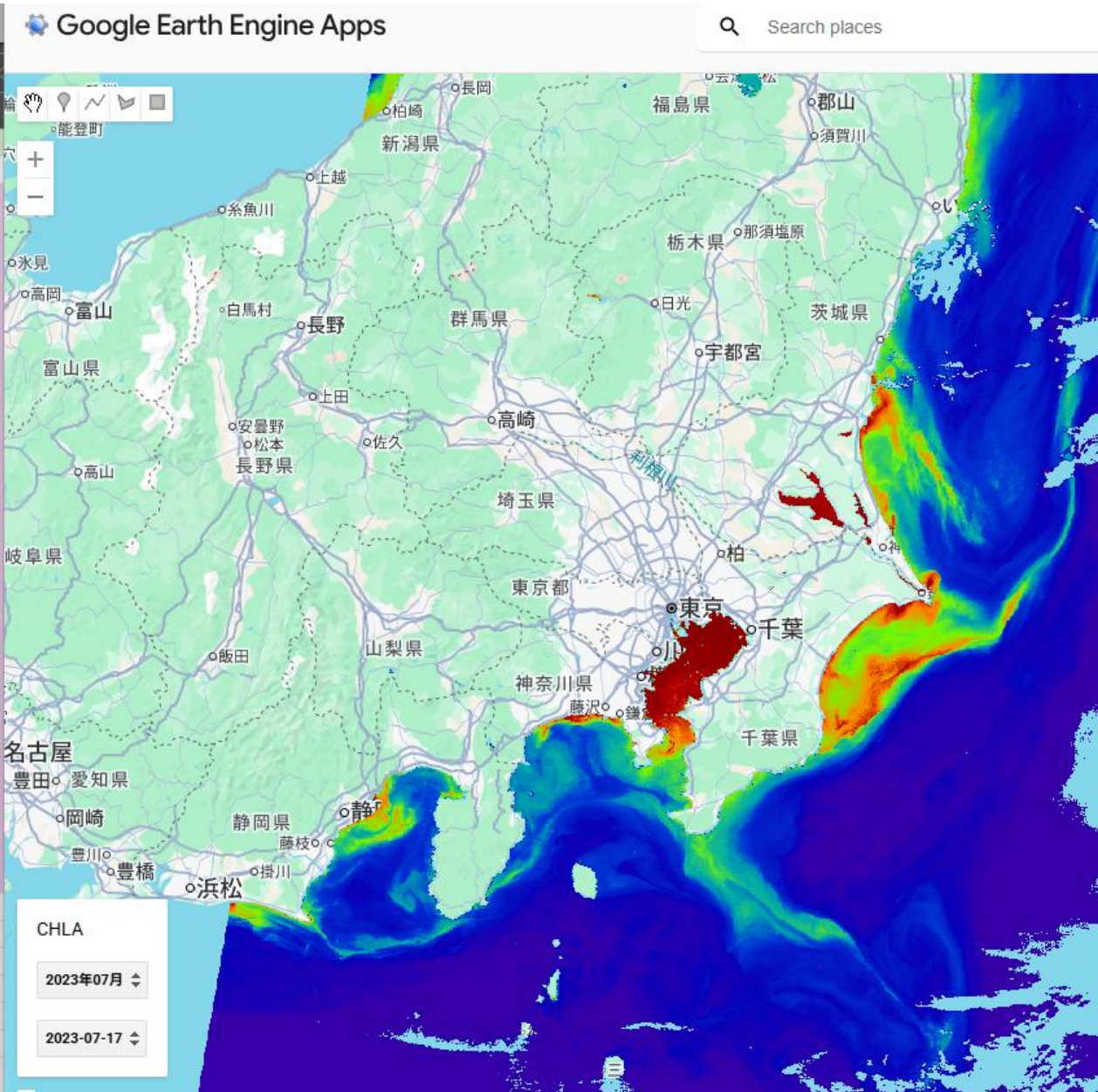
Land Surface Temperature (Daytime/Monthly)

Land surface temperature product in GCOM-C SGLI Level-3 Standard product Version 3.

ID:

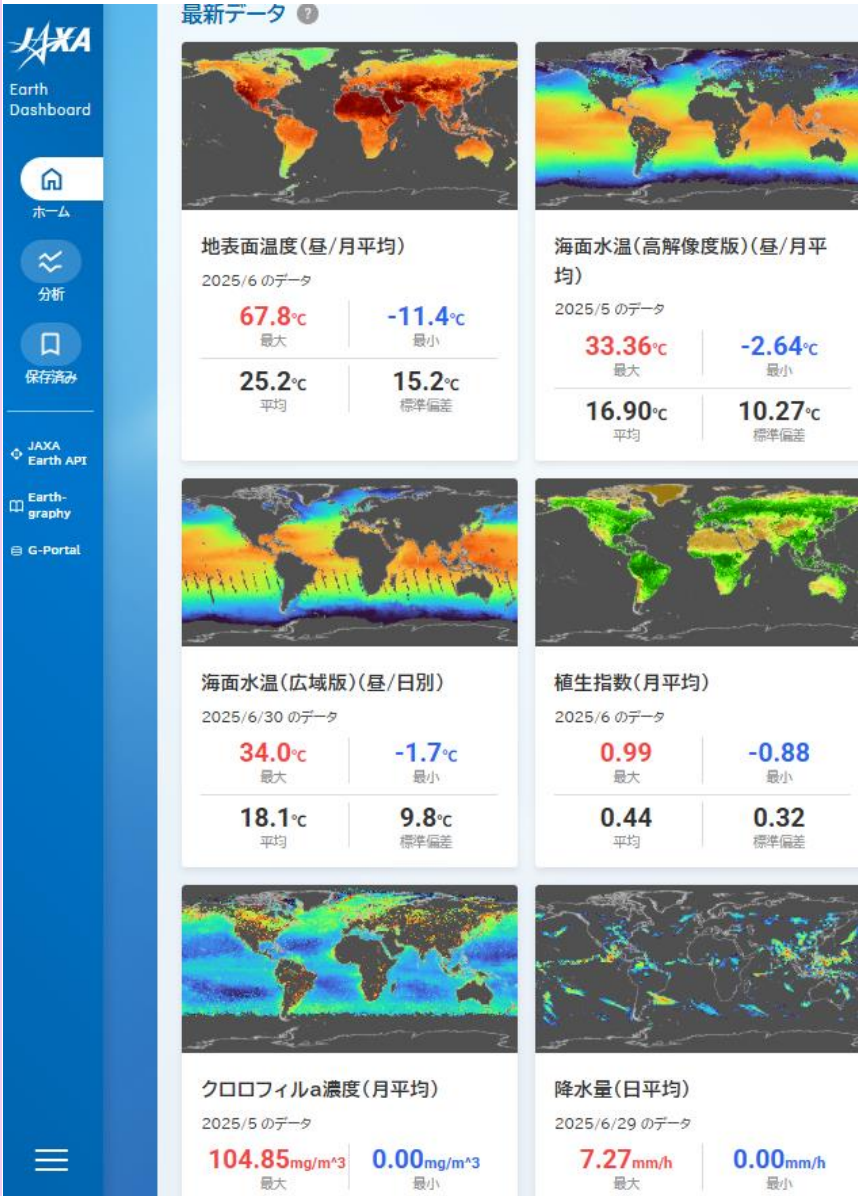
BAND:

Visualization



- Visualize the SGLI data on GEE
- Japan area is available
- RGB, LST, CHLA, TSM, SST, FAI, RSRF

JAXA Earth Dashboard



- Visualize the JAXA's dataset like a dashboard (informative?)

分析したいデータを選択する

観測対象から探す

カテゴリから探す

林野火災

☐ 干ばつ指数(月平均) ?

☐ 土壌水分量(昼/月平均) ?

☐ 土壌水分量(昼/半月平均) ?

☐ 土壌水分量(昼/日別) ?

☐ 土壌水分量(夜/月平均) ?

☐ 土壌水分量(夜/半月平均) ?

☐ 土壌水分量(夜/日別) ?

☐ エアロゾル量(月平均) ?

☐ エアロゾル量(半月平均) ?

☐ エアロゾル量(日別) ?

大雨

☐ 降水量(月平均) ?

☐ 降水量(半月平均) ?

☐ 降水量(日平均) ?

☐ 干ばつ指数(月平均) ?

☐ 土壌水分量(昼/月平均) ?

☐ 土壌水分量(昼/半月平均) ?

☐ 土壌水分量(昼/日別) ?

☐ 土壌水分量(夜/月平均) ?

☐ 土壌水分量(夜/半月平均) ?

☐ 土壌水分量(夜/日別) ?

台風

☐ 海面水温(広域版)(昼/日別) ?

☐ 海面水温(広域版)(夜/日別) ?

☐ 降水量(月平均) ?

☐ 降水量(半月平均) ?

☐ 降水量(日平均) ?

漁業

☐ 海面水温(高解像度版)(昼/月平均) ?

☐ 海面水温(高解像度版)(昼/半月平均) ?

☐ 海面水温(高解像度版)(昼/日別) ?

分析する